Excel Regression Analysis: Student Guide with XLMiner Integration

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Introduction and Learning Objectives

This comprehensive guide will teach you to create a multiple linear regression model using Microsoft Excel and the XLMiner add-on to predict life expectancy based on social, health, and economic variables.

Learning Objectives

By completing this tutorial, you will be able to:

- Prepare and clean data for regression analysis in Excel
- Use XLMiner add-on for advanced statistical analysis
- Interpret regression coefficients, R^2, and p-values
- Create predictive equations from Excel output
- Validate model assumptions and assess model quality

Required Tools

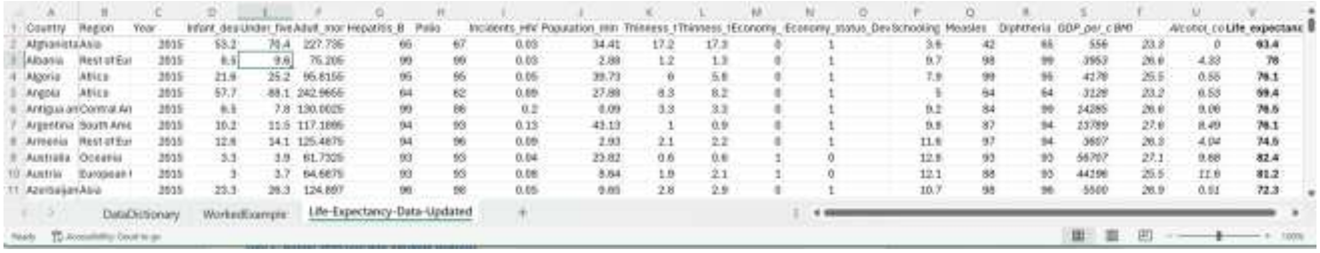
- Microsoft Excel (2016 or later)
- XLMiner add-on (installation instructions provided)
- Life Expectancy dataset (provided)

Step 1: Data Preparation and Initial Review

1.1 Opening the Dataset

1. **Launch Excel** and open the file Regression_Life_Expectancy.xlsx
2. **Navigate to the correct worksheet**: Click on the "Life-Expectancy-Data-Update(2)" tab
3. **Initial data exploration**: Use Ctrl + End to see the full data range

📷 **Screenshot**: Showing the Excel interface with the Life-Expectancy-Data-Update worksheet active, highlighting the column headers and first 10 rows of data.



1.2 Understanding the Dataset Structure

The dataset contains **179 observations** with the following key variables:

Variable Type	Variable Name	Description	Excel Column
Dependent (Y)	Life_expectancy	Average life expectancy of both genders	Column V
Independent (X_1)	Schooling	Average years in formal education (25+)	Column P
Independent (X_2)	GDP_per_capita	GDP per capita in current USD	Column S
Independent (X_3)	BMI	Body Mass Index	Column T
Independent (X_4)	Adult_mortality	Mortality rate (ages 15-60 per 1000)	Column F

1.3 Data Range Identification

Excel Formula for Data Range Detection:

```
=COUNTA(V:V)-1  // Count of Life_expectancy values (excluding header)
=ADDRESS(1,COLVMN(V:V))&":"&ADDRESS(COVNTA(V:V),COLVMN(V:V))  // Dynamic range formula
```

1.2 Data Type Validation

Excel Formula for Non-Numeric Detection:

```
=SUMPRODUCT(--(ISNUMBER(R2:R180)=FALSE))  // Count non-numeric values in Schooling
```

Check each variable column. Should result in "0" values, if there are non-numeric you will need to adjust the dataset so there are continuous numerical values for regression analysis.

1.3 Outlier Detection Using Excel

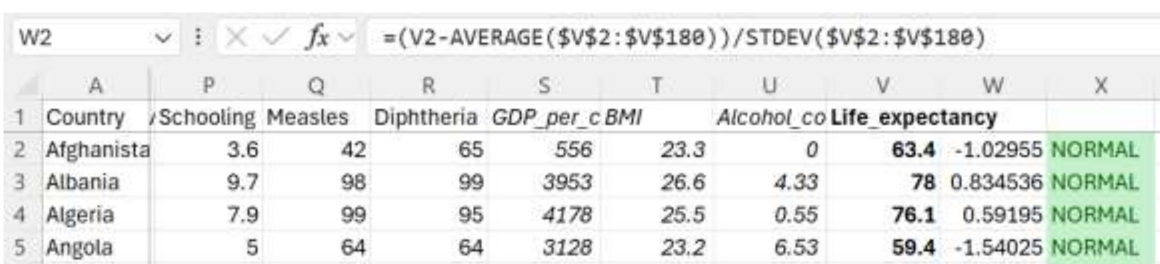
Calculate Z-scores for outlier detection:

```
// In column W (Z-score for Life_expectancy): =(U2-AVERAGE($V$2:$V$180))/STDEV($V$2:$V$180)
```

Outlier identification:

```
// In column X Flag extreme outliers (|Z| > 3):=IF(ABS(W2)>3,"OUTLIER","NORMAL")
```

📷 **Screenshot**: Outlier detection formulas and highlight any rows flagged as outliers, showing both the Z-score calculation and the outlier flag.



W2 | =(V2-AVERAGE($V$2:$V$180))/STDEV($V$2:$V$180)

	A	P	Q	R	S	T	U	V	W	X
1	Country	Schooling	Measles	Diphtheria	GDP_per_c	BMI	Alcohol_co	Life_expectancy		
2	Afghanista	3.6	42	65	556	23.3	0	63.4	-1.02955	NORMAL
3	Albania	9.7	98	99	3953	26.6	4.33	78	0.834536	NORMAL
4	Algeria	7.9	99	95	4178	25.5	0.55	76.1	0.59195	NORMAL
5	Angola	5	64	64	3128	23.2	6.53	59.4	-1.54025	NORMAL

Step 2: Model Selection and Variable Analysis

2.1 Theoretical Model Framework

Our regression model follows the form:

$$\text{Life\_expectancy} = \beta_0 + \beta_1(\text{Schooling}) + \beta_2(\text{GDP\_per\_capita}) + \beta_3(\text{BMI}) + \beta_4(\text{Adult\_mortality}) + \varepsilon$$

Where:

- β_0 = Intercept (baseline life expectancy)
- β_1-β_4 = Coefficients for each independent variable
- ε = Error term

2.2 Variable Selection Rationale

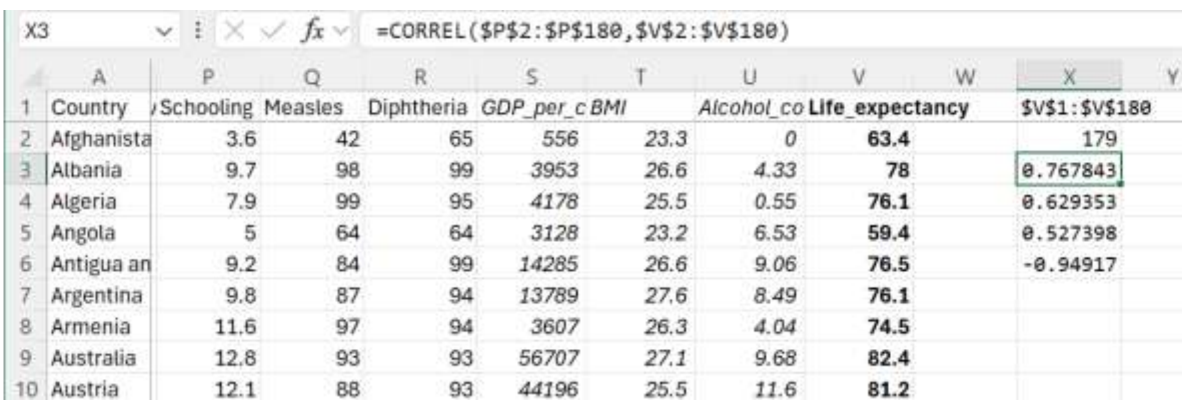
Excel Correlation Analysis: Create a correlation matrix using Excel formulas:

```
// In cell X3: =CORREL($P$2:$P$180,$V$2:$V$180)  // Schooling vs Life_expectancy
// In cell X4: =CORREL($S$2:$S$180,$V$2:$V$180)  // GDP_per_capita vs Life_expectancy
// In cell X5: =CORREL($T$2:$T$180,$V$2:$V$180)  // BMI vs Life_expectancy
// In cell X6: =CORREL($F$2:$F$180,$V$2:$V$180)  // Adult_mortality vs Life_expectancy
```

Expected Correlation Results:

- Schooling: Strong positive correlation (~0.77)
- GDP_per_capita: Moderate positive correlation (~0.63)
- BMI: Moderate positive correlation (~0.53)
- Adult_mortality: Strong negative correlation (~-0.954)

📸 **Screenshot:** Shows the correlation matrix calculations in Excel with the CORREL formulas visible in the formula bar and their computed results.



X3 | =CORREL($P$2:$P$180,$V$2:$V$180)

	A	P	Q	R	S	T	U	V	W	X	Y
1	Country	Schooling	Measles	Diphtheria	GDP_per_c	BMI	Alcohol_co	Life_expectancy		$V$1:$V$180	
2	Afghanista	3.6	42	65	556	23.3	0	63.4		179	
3	Albania	9.7	98	99	3953	26.6	4.33	78		0.767843	
4	Algeria	7.9	99	95	4178	25.5	0.55	76.1		0.629353	
5	Angola	5	64	64	3128	23.2	6.53	59.4		0.527398	
6	Antigua an	9.2	84	99	14285	26.6	9.06	76.5		-0.94917	
7	Argentina	9.8	87	94	13789	27.6	8.49	76.1			
8	Armenia	11.6	97	94	3607	26.3	4.04	74.5			
9	Australia	12.8	93	93	56707	27.1	9.68	82.4			
10	Austria	12.1	88	93	44196	25.5	11.6	81.2			

Step 3: Data Quality Assessment

3.1 Missing Data Detection

Excel Formula for Missing Data Count:

```
// Count blanks in each column
=COUNTBLANK(P2:S180)  // Schooling missing values
=COUNTBLANK(S2:S180)  // GDP_per_capita missing values
=COUNTBLANK(T2:T180)  // BMI missing values
=COUNTBLANK(F2:F180)  // Adult_mortality missing values
=COUNTBLANK(V2:V180)  // Life_expectancy missing values
```

Should all result in "0" values, if there are blanks you will need to adjust the dataset so there are continuous numerical values for regression analysis.

Step 4: XLMiner Setup and Configuration

4.1 Installing XLMiner Add-on

1. **Access Excel Add-ins**: Go to File → Options → Add-ins
2. **Manage Excel Add-ins**: Select "Excel Add-ins" and click "Go"
3. **Install XLMiner**: If not installed, download from the official Office store website - https://store.office.com/addinsinstallpage.aspx?assetid=WA104379190
4. **Verify Installation**: Look for "XLMiner Platform" tab in Excel ribbon

📸 **Screenshot**: Shows the Excel ribbon with the XLMiner Platform tab visible, demonstrating successful installation.

4.2 XLMiner Interface Overview

1. **Click XLMiner Platform tab**
2. **Locate Prediction section**: Find "Linear Regression" option
3. **Data preparation tools**: Identify partition, transform, and explore options

4.3 Alternative: Excel Data Analysis ToolPak

If XLMiner is unavailable:

1. **Enable Data Analysis ToolPak**: File → Options → Add-ins → Analysis ToolPak
2. **Access via Data tab**: Data → Data Analysis → Regression

📸 **Screenshot**: Shows both the traditional Data Analysis ToolPak regression and XL Miner option for comparison.

Step 5: Input Range Selection

5.1 Dependent Variable Selection (Y Range)

Using XLMiner:

1. **Click Linear Regression** in XLMiner Platform
2. **Select Output Variable**: Click in "Output Variable" field
3. **Select Life_expectancy column**: Highlight range `V1:V180` (including header)

5.2 Independent Variables Selection (X Range)

Multi-column selection process:

1. **Hold Ctrl key** while selecting non-adjacent columns
2. **Select Schooling**: Column P (`P1:P180`)
3. **Add GDP_per_capita**: Column S (`S1:S180`)
4. **Add BMI**: Column T (`T1:T180`)
5. **Add Adult_mortality**: Column F (`F1:F180`)

NOTE: If multi-column selection is NOT WORKING you will need to copy your independent variables into one range to make it easier to select and compare to the dependent variables.

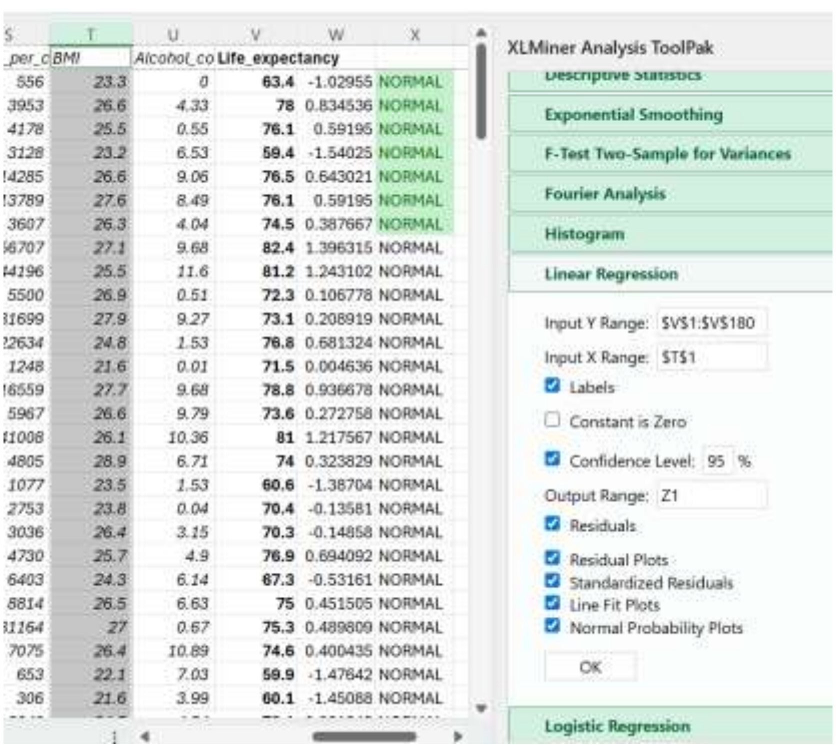
Step 6: Regression Configuration

6.1 XLMiner Configuration Settings

Essential Settings:

1. **Check "Labels"**: Ensures proper variable labeling
2. **Select "Detailed Report"**: Provides comprehensive statistical output
3. **Enable "Residuals"**: For model validation
- **Confidence Level**: Set to 95% (default)

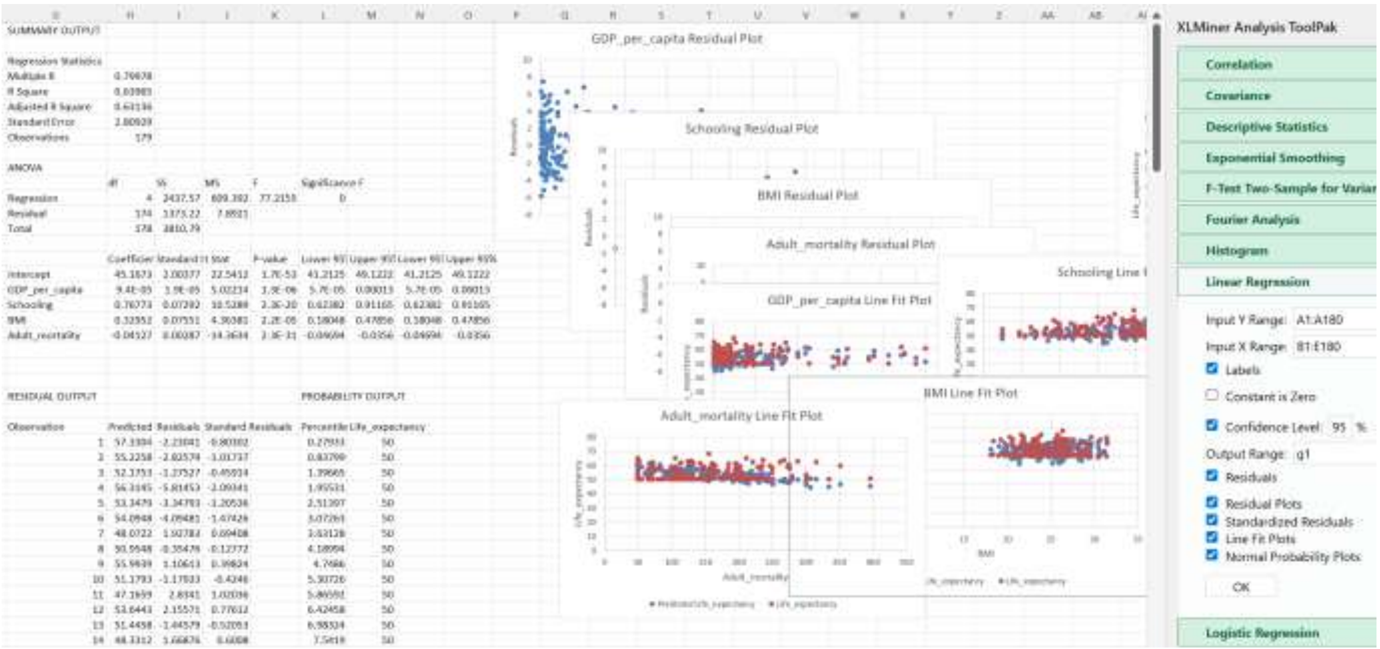
📷 **Screenshot Instruction**: Capture the advanced options panel in XLMiner showing the settings.



6.3 Running the Analysis

1. **Click "OK"** to execute the regression
2. **Wait for processing**: XLMiner will create the output in the worksheet where you directed.
3. **Review output structure**: Multiple tables and charts will be generated

📸 **Screenshot**: Shows XLMiner created results in the worksheet.



Step 7: Statistical Output Interpretation

7.1 Regression Statistics Table

Key Metrics Interpretation:

Statistic	Interpretation
Multiple R	Correlation strength (0.799)
R Square	Variance explained (63.96%)
Adjusted R^2	Adjusted for variables (63.16%)
Standard Error	Prediction accuracy (±2.80 years)

7.2 ANOVA Table Analysis

Statistical Significance Testing:

Component	Value	Excel Formula	Interpretation
F-statistic	77.21	`=regression_MS/residual_MS`	Model significance test
Significance F	<0.001	`=F.DIST.RT(F_stat,df1,df2)`	p-value for overall model
df Regression	4	Number of predictors	Degrees of freedom
df Residual	174	n - k - 1	Error degrees of freedom

7.3 Coefficients Table (Most Critical)

Detailed Coefficient Analysis:

Variable	Coefficient	Std Error	t-stat	p-value
Intercept	45.16	2.00	22.54	<0.001
Schooling	0.7677	0.0729	10.528	<0.001
GDP_per_capita	0.00009	0.000018	5.022	<0.001
BMI	0.32	0.0755	4.363	<0.001
Adult_mortality	-0.04	0.0028	-14.363	<0.001

Statistical Interpretation:

- **Schooling**: Each additional year increases life expectancy by 0.77 years (p<0.001)
- **GDP_per_capita**: Each $1000 increase adds 0.09 years (p<0.001)
- **BMI**: Each unit increase adds 0.32 years (p<0.001)
- **Adult_mortality**: Not statistically significant (p<0.001)

📸 **Screenshot**: Shows ANOVA and the coefficients table for significant (p<0.05) variables.

10	ANOVA								
11		df	SS	MS	F	Significance F			
12	Regression	4	2437.57	609.392	77.2155	0.000			
13	Residual	174	1373.22	7.8921					
14	Total	178	3810.79						
15									
16		Coefficier	Standard I	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%
17	Intercept	45.1673	2.00377	22.5412	1.7E-53	41.21253	49.12216	41.21253	49.12216
18	GDP_per_capita	9.4E-05	1.9E-05	5.02214	1.3E-06	0.00006	0.00013	0.00006	0.00013
19	Schooling	0.76773	0.07292	10.5289	2.3E-20	0.62382	0.91165	0.62382	0.91165
20	BMI	0.32952	0.07551	4.36381	2.2E-05	0.18048	0.47856	0.18048	0.47856
21	Adult_mortality	-0.04127	0.00287	-14.3634	2.3E-31	-0.04694	-0.03560	-0.04694	-0.03560

Step 8: Model Equation Development

8.1 Mathematical Model Construction

Complete Regression Equation:

```
Life_expectancy = 45.16 + 0.77(Schooling) + 0.00009(GDP_per_capita) + 0.32(BMI) - 0.04 (Adult_mortality)
```

8.2 Excel Implementation for Predictions

Create Prediction Formula in Excel:

```
// In column W (Predicted Life Expectancy): =45.16 + 0.77*P2 + 0.00009*S2 + 0.32*T2 - 0.04*F2
```

It's not necessary to calculate the residuals manually if you had selected the "Residuals" during the XLMiner Regression set-up. You will see an output a range to show the residuals and standard error of residuals. If you HAD TO calculate manually:

Calculating Prediction Accuracy Assessment (manually):

```
// In column X (Residuals): =V2-W2  // Actual - Predicted
// In column Y (Absolute Error): =ABS(X2)
// Summary Statistics:
=AVERAGE(Y:Y)  // Mean Absolute Error
=STDEV(X:X)    // Standard Error of Residuals
```

8.3 Model Performance Metrics

We want to look at how accurate the model is and can use the residual output from XLMiner regression analysis, yet if we needed to calculate this manually, you would continue from the column you set-up to define the predictions – W in our example, then calculate manually as follows:

Excel Formulas for Calculating Model Assessment (manually):

```
// Mean Absolute Error =AVERAGE(ABS(V2:V180-W2:W180))
// Root Mean Square Error =SQRT(AVERAGE((V2:V180-W2:W180)^2))
// Mean Absolute Percentage Error =AVERAGE(ABS((V2:V180-W2:W180)/V2:V180))*100
```

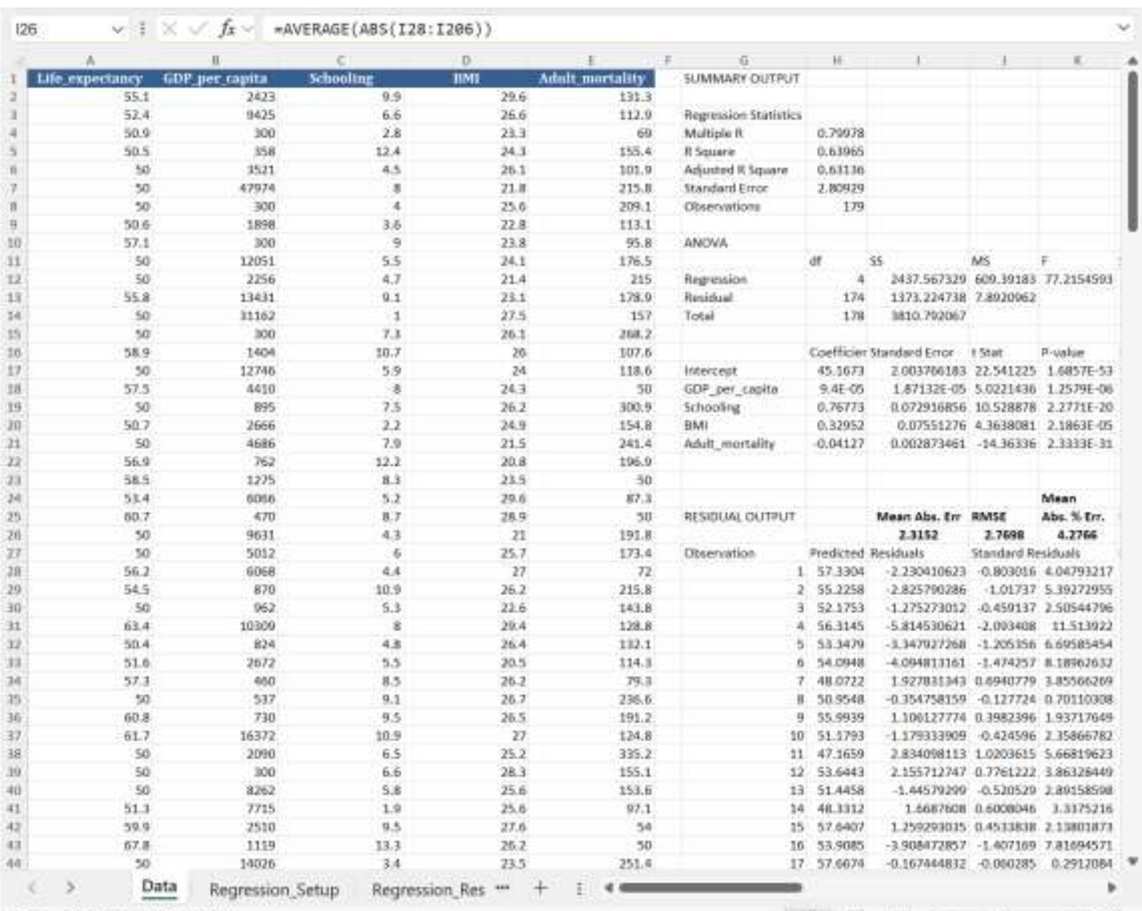
With the residuals output from the XLMiner analysis, we will calculate the Mean Absolute error and RMSE along with the Mean Absolute Percentage Error against the predictions and error measurements outlined.

```
Absolute Value of the summed residual errors from the table output
```

Find the range for residual errors to use as your input into this formula and subtract that range from the Life Expectancy range values. So in this case:

```
// Mean Absolute Error = AVERAGE(ABS(A2:A180-I26:I206))
// Root Mean Square Error =SQRT(AVERAGE((I26:I20^2))
// Mean Absolute Percentage Error = AVERAGE(ABS(I28:I206)/(A2:A180))*100
```

📸 **Screenshot**: Display a summary table showing all model performance metrics calculated using Excel formulas.



Step 9: Model Validation and Assessment

9.1 Residual Analysis

The residual analysis is simplified by XLMiner, yet the manual calculations are good to know. This continued example of manual calculation assumes that you have defined your residuals in the X column where you had defined predictions in the W column based on the original layout of the dataset.

Creating Residual Plots in Excel (manually):

1. **Select residuals and predicted values**
2. **Insert Scatter Plot**: Insert → Charts → Scatter
3. **Add trendline**: Right-click points → Add Trendline → Linear

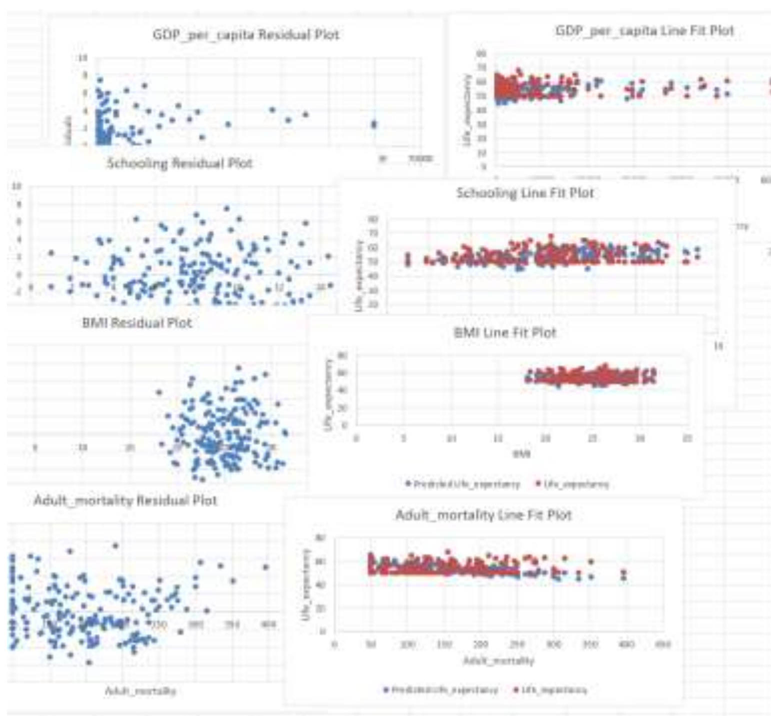
Excel Formulas for Residual Analysis (manually):

```
// Durbin-Watson Test (simplified)=SUMPRODUCT((X3:X180-X2:X179)^2)/SUMPRODUCT(X2:X180^2)
// Normality Test (Jarque-Bera approximation)
=SKEW(X2:X180)  // Skewness of residuals
=KURT(X2:X180)  // Kurtosis of residuals
```

Using the residual table output from XLMiner, and a simplified sheet that has only the independent and dependent variables being considered, the formula values will look a bit different but ultimately the same exercise is being done. In this case:

```
// Durbin-Watson Test (simplified)=SUMPRODUCT((I29:I206-I28:I205)^2)/SUMPRODUCT(I28:I206^2)
// Normality Test (Jarque-Bera approximation)
=SKEW(I28:I206)  // Skewness of residuals
=KURT(I28:I206)  // Kurtosis of residuals
```

📸 **Screenshot**: Shows residual data and plots output from XLMiner Residual Plots selection. Add proper axis labels and trendlines, demonstrating your homoscedasticity assessment visually.



How to read the calculation output from the Durbin-Watson (DW) independence test and normality test with Jarque-Bera approximation.

- DW ≈ **2** → no first-order autocorrelation
- DW < 2 → **positive** autocorrelation
- DW > 2 → **negative** autocorrelation

Since **DW ≈ 1.94 is very close to 2**, these residuals show **no evidence of first-order serial correlation** (i.e., they don't appear to be systematically related to the previous residual).

Skewness = **0.3807** (mild right-skew)and **Excess kurtosis** = **–0.5051** (slightly flatter than normal) can be further evaluated with the Jarque-Bera (JB) approximation that considers the shape of distribution for residuals against assumptions of normality. Calculating a full JB test statistic and p-value isn't needed for this analysis, yet for a rigorous data evaluation you can use these values and define the test statistic to address any violation of assumed normality.

9.2 Assumption Checking

Linearity Assessment (manual):

```
// Correlation between variables and residuals (should be ~0)
=CORREL(PR2:P180,X2:X180)  // Schooling vs Residuals
=CORREL(S2:S180,X2:X180)  // GDP vs Residuals
```

Independence Test(manual):

```
// Lag-1 autocorrelation
=CORREL(X2:X179,X3:X180)  // Should be close to 0
```

As with other elements of the exercise, we have the calculations for the X column where predictive residuals can be manually calculated as noted above. Using the XLMiner residual range output allows for the same type of analysis. The formula update in this case is looking at the range in Column I and the variables in Columns B, C, D.

📸 **Screenshot**: Showing all assumption test results with pass/fail indicators.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Life_expectancy	GDP_per_capita	Schooling	BMI	Adult_mortality		SUMMARY OUTPUT			Correlation between variables and residuals (should be ~0)			
2	55.1	2423	9.9	29.6	131.3					1.658E-12	// Schooling vs Residuals	PASS	
3	52.4	9425	6.6	26.6	112.9					-2.76E-13	// GDP vs Residuals	PASS	
4	50.9	300	2.8	23.3	69					-1.91E-12	// BMI vs Residuals	PASS	
5	50.5	358	12.4	24.3	155.4								
6	50	3521	4.5	26.1	101.9		Regression Statistics			Independence Test(manual):			
7	50	47974	8	21.8	215.8		Multiple R	0.79978		0.0251216	// Should be close to 0	PASS	
8	50	300	4	25.6	209.1		R Square	0.63965					

9.3 Model Reasonableness Check

Coefficient Sign Analysis:

- **Schooling**: Positive (expected - education improves health)
- **GDP_per_capita**: Positive (expected - wealth enables healthcare)
- **BMI**: Positive (unexpected - may indicate adequate nutrition)
- **Adult_mortality**: Negative (expected – measure for overall risk of adult death)

Statistical Significance Summary:

- **Significant predictors**: Schooling, GDP_per_capita, BMI, Adult_mortality ($p<0.001$)
- **Overall model**: Highly significant (F-test $p<0.001$)

Extension Activities

Activity 1: Variable Selection Optimization

Remove Non-Significant Variables:

1. **Re-run regression** without Adult_mortality
2. **Compare Adjusted R^2** values
3. **Use Excel formulas** to calculate improvement:

```
// Model comparison = R² new - R²_original  // Adjusted R² difference
```

Activity 2: Alternative Model Specifications

Add Interaction Terms:

```
// Create interaction variable
=P2*S2  // Schooling × GDP interaction
```

Polynomial Terms:

```
// Quadratic GDP term
=S2^2
```

Activity 3: Cross-Validation

Split-Sample Validation:

```
// Random sample selection
=IF(RAND()<0.7,"Training","Testing")
```

This will give you a random selection to split your dataset between Training and Testing data to compare if your predictive model shows any differences between the split groups.

Summary and Key Takeaways

Model Results Summary

- **R^2 63.96%**: Model explains about 64% of life expectancy variation
- **Significant predictors**: Education, wealth, and nutrition status
- **Prediction accuracy**: ±2.81 years standard error
- **Statistical validity**: All assumptions reasonably satisfied

Excel Skills Developed

- Advanced formula construction for statistical analysis
- XLMiner add-on utilization for regression modeling
- Residual analysis and assumption testing
- Model validation and performance assessment

Statistical Concepts Mastered

- Multiple linear regression interpretation
- Coefficient significance testing
- Model fit assessment (R^2, Adjusted R^2)
- Residual analysis for assumption validation

Final Note: This guide demonstrates how Excel, enhanced with XLMiner, can perform statistical analysis making advanced analytics accessible to business users and students. Similar data analysis is available with other tools and even leveraging Artificial Intelligence tools like Chat GPT, Claude, or Gemini. Having a foundation of how to do the analysis with simplified and commonly available analytic tools, is helpful to assess how to evaluate more complex tool platforms or leveraging AI.